

Chemistry - Graduate Level Assessment 11

Instructions

- Answer all questions.

Multiple Choice

Question 1

(2 marks)

What is the approximate boiling point at standard pressure of a solution prepared by dissolving 234 g of NaCl in 500 g of H₂O?

- A. 102.5[°]C
- B. 100[°]C
- C. 109.5[°]C
- D. 106.78[°]C
- E. 101.22[°]C

Question 2

(2 marks)

A substance that is NOT generally considered to be a toxic pollutant in water is

- A. nitrogen
- B. a halogenated hydrocarbon
- C. chlorine
- D. copper
- E. arsenic

Question 3

(2 marks)

Of the following compounds, which has the lowest melting point?

A. NaCl

B. LiCl

C. AlCl₃

D. KCl

E. FeCl₃

Question 4

(2 marks)

10.0 grams of urea, CO(NH₂)₂, is dissolved in 200 grams of water. What are the boiling and freezing points of the resulting solution?

A. Boiling point is 101.00°C, Freezing point is 1.55°C

B. Boiling point is 101.00°C, Freezing point is -2.00°C

C. Boiling point is 99.57°C, Freezing point is 1.55°C

D. Boiling point is 100.0°C, Freezing point is 0.0°C

E. Boiling point is 100.43°C, Freezing point is 1.55°C

Question 5

(2 marks)

Calculate the weight in grams of sulfuric acid in 2.00 liters of 0.100 molar solution. (MW of H₂SO₄ = 98.1)

A. 10.0 g

B. 24.4 g

C. 15.2 g

D. 22.3 g

E. 19.6 g

True / False

Question 1

(2 marks)

Compounds like Na_2O and CaCl_2 also require Roman numerals in their names due to their ionic nature.

True

False

Question 2

(2 marks)

Diluting 50 ml of 3.50 M H_2SO_4 to achieve a concentration of 2.00 M requires a final volume of 125 ml.

True

False

Question 3

(2 marks)

The quotient $\frac{2 \times 10^0}{(2 \times 10^0)}$ simplifies to 1, since both the numerator and denominator evaluate to the same value.

True

False

Question 4

(2 marks)

To produce 1.50 g of O_2 , 8.45 g of NaNO_3 must be decomposed according to the reaction.

True

False

Question 5

(2 marks)

The zero-point entropy of NO is $0.00 \text{ cal mol}^{-1} \text{ K}^{-1}$, indicating no disorder at absolute zero.

True

False

Long Form

Question 1

(10 marks)

Calculate the cell potential E for the $\text{Ag}/\text{AgCl}/\text{Cl}^-$ ($a = 0.0001$) and Ce^{4+} ($a = 0.01$), Ce^{3+} ($a = 0.1$)/Pt system, discussing the significance of standard half-cell potentials in your analysis.

Type your answer here...

Question 2

(10 marks)

Discuss the stoichiometric principles involved in the decomposition of mercury(II) oxide (HgO) to form oxygen gas (O_2) and liquid mercury (Hg). Calculate the moles of oxygen produced from 4.32 g of HgO , given its molar mass is 216 g/mol.

Type your answer here...

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